

Advanced control strategies for continuous capture of monoclonal antibodies based upon biolayer interferometry

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Abstract

The semi and fully continuous production of monoclonal antibodies (mAbs) has been gaining traction as a lower cost, and efficient production of mAbs to broaden patient access. To be truly flexible and adaptive to process demands, the industry has lacked sufficient advanced control strategies. The variation of the upstream product concentration typically cannot be handled by the downstream capture step, which is configured for a constant feed concentration and fixed binding capacity. This inflexibility leads to losses of efficiency and product yield.

This study shows that these challenges can be overcome by a novel advanced control strategy concept that includes dynamic control throughout a perfusion bioreactor, with cell retention by alternating tangential flow, integrated with simulated moving bed (SMB) multi-column chromatography. The automation workflow and advanced control strategy were implemented through the use of a visual programming development environment. This enabled dynamic flow control across the upstream and downstream process integrated with a dynamic column loading of the SMB. A sensor prototype, based on continuous biolayer interferometry measurements was applied to detect mAb breakthrough within the last column flow-through to manage column switching. This novel approach provided higher specificity and lower background signal compared to commonly used spectroscopy methods, resulting in an optimized resin utilization while simultaneously avoiding product loss. The dynamic loading was found to provide a twofold increase of the mAb concentration in the eluate compared to a conservative approach with a predefined recipe with similar impurity removal. This concept shows that advanced control strategies can lead to significant process efficiency and yield improvement.

KEYWORDS

advanced control strategies, biolayer interferometry, continuous processes, mAb, PAT, process intensification

1 | INTRODUCTION

Monoclonal antibodies (mAbs) are used for treatment of numerous severe diseases like cancer, infectious autoimmune disorders, and inflammatory diseases (Andreaskos et al., 2002; Kesik-Brodacka, 2018; Reichert et al., 2005; Santos et al., 2018). Due to the high specificity, activity and fewer side effects compared to conventional drugs (Kesik-Brodacka, 2018; Mitragotri et al., 2014) the market demand for mAbs continues to rise, representing over 60% of the global biopharmaceutical industry revenue (Levine & Cooney, 2017).

To broaden patient access, efforts continue to improve manufacturing efficiency and reduce costs. To this end process intensification with semi and fully continuous biomanufacturing is being pursued and implemented into clinical production (Bielser et al., 2018; Godawat et al., 2015; Xu et al., 2020). For continuous upstream process (USP) cultivations, perfusion operations are commonly used. These require the continuous addition of unspent media, removal of the spent media containing the mAb by permeate withdrawal and the retaining of the cells in the bioreactor. As cell retention devices alternating tangential flow (ATF) or tangential flow filtration devices, centrifuges or settlers can be used (Chotteau et al., 2014; Müller et al., 2022; Subramanian, 2014; Voisard et al., 2003). Despite the several benefits of continuous perfusion operation, that include increased volumetric productivity (Müller et al., 2022), the implementation in the highly regulated biomanufacturing industry remains challenging due to the complexity of the process development and the required automated control strategy (Croughan et al., 2015; Kelley et al., 2018; Schwarz et al., 2022). Tools to ease implementation and execution will help to widen industry adoption for mAb production (Croughan et al., 2015).

To fully exploit the possible benefits of such a continuous USP, an intensified downstream process (DSP) is required. The routinely applied protein A affinity chromatography step as first mAb capture, offers great potential for improvement due to its limited loading capacity and high resin cost (Shukla et al., 2007). As mitigating strategy many companies cycle the capture columns several times during a batch, resulting in a smaller volume of the resin required for the same amount of mAb. However, reduced throughput, longer processing times and related potential impacts on the product quality are issues of concern (Bisschops & Brower, 2013).

The application of continuous simulated moving bed (SMB) chromatography as an approach for DSP intensification offers an increased productivity with simultaneously reduced buffer consumption and manufacturing facility footprint, while maintaining product yield and purity (Angarita et al., 2015; Baur et al., 2016; Bisschops & Brower, 2013). The implementation of SMB technology allows the resin particles to remain in a packed bed while the "transport" of the resin is realized by means of switching column in- and outlets. By the usage of a multi-column chromatography approach product breakthrough of one column is loaded on another one, avoiding product loss while simultaneously maximal resin utilization, exceeding the dynamic binding capacity, can be obtained. This allows for significant efficiency improvements, as large batch chromatography columns are

often loaded to only 65% of their static binding capacity (Bisschops & Brower, 2013). Once fully loaded a column switch is applied and the column is washed, eluted, and regenerated, before entering the next loading cycle. Using a countercurrent process, the efficiency is further improved due to an optimization of the driving force for mass transfer throughout the entire course of the contact zone. As a result, the mass transfer exceeds the thermodynamic limitations of a batch or concurrent chromatography process (Bisschops & Brower, 2013).

Despite its numerous advantages continuous SMB processes commonly rely on predefined recipes with regard to loading volumes and flow rates and consequently for the column loading time, which is also determining for the other non-loading steps to ensure that the column can go through all the steps of the recipe. This requires several assumptions like a constant and known product concentration in the feed solution, that all columns have the same binding capacity and that the binding capacity remains constant throughout the process (Rolinger et al., 2021). Deviations from these assumptions result in either a low yield due to product loss or, in the case of overestimated product concentration, in less efficient processes with a low productivity. These challenges could be overcome by a dynamic loading based on breakthrough detection of the mAb within the flow-through of the last column in the loading cycle, which then triggers the column switch. This offers several benefits like the ability to compensate for variable mAb concentrations in the feed solutions (e.g., by variations in the cell cultivation) and a minimization of product loss due to variations in the column capacity.

To perform (near) real-time process monitoring and control, several process analytical technology (PAT) tools have been developed for fast mAb quantification, like infrared (Capito et al., 2015; Thakur et al., 2020) or Raman spectroscopy (Feidl et al., 2019; Rolinger et al., 2021). However, most commonly UV spectroscopy is used for mAb breakthrough detection. One approach is the calculation of a difference signal of two detectors, situated at the column inlet and the column outlet using the UV absorption at 280 nm (Warikoo et al., 2012). Other studies implemented partial least squares regression modeling on UV/Vis absorption spectra instead of single wavelength measurements (Brestrich et al., 2015; Rüdts et al., 2017). However, these methods exhibit several disadvantages like the low specificity of the UV signal (mAb and impurities contribute to the absorption at 280 nm) and the very high background signal due to the media components in the feed solution (Gupta, 2002). This poses challenges to detect a very low mAb concentration within the flow-through to avoid product loss.

A promising method to overcome these limitations is biolayer interferometry (BLI). This label-free technology is based on the interference pattern obtained from the combination of white light reflected from an internal reference surface and a biolayer (Do et al., 2008; Kumaraswamy & Tobias, 2015; Müller-Esparza et al., 2020; Schasfoort & Tudos, 2008). The high affinity and consequently binding of the analyte to the immobilized molecule at the biocompatible surface result in an increased thickness at the surface of the biosensor (Figure 1). This in turn is measured as concentration dependent signal by the change in the light

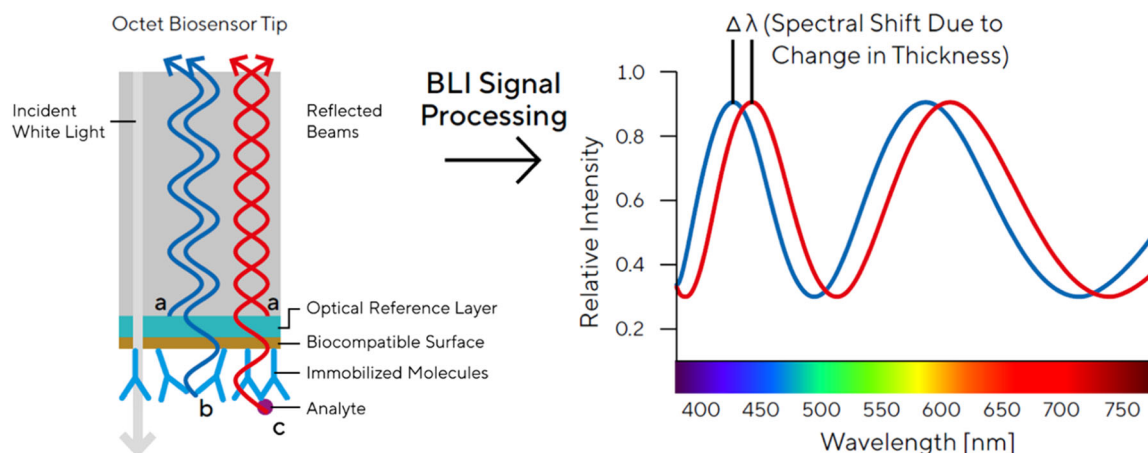


FIGURE 1 An optical fiber used for biolayer interferometry and a typical optical signal. White light shines down the biosensor and a small portion of the light is reflected back along the fiber. The reflected light originates from two interfaces: (a) the optical reference layer and (b) the interface between the immobilized protein and the surrounding solution (blue reflected beams). When an analyte in the solution binds to the end of the biosensor, the second reflective interface is shifted (c). This changes the interference pattern (red reflected beams) resulting in a spectral shift. Reproduced from Do et al. (2008).

interference (Do et al., 2008; Kumaraswamy & Tobias, 2015; Müller-Esparza et al., 2020). For the quantification of mAbs, specific biosensors with immobilized protein A on their surface can be used.

The BLI technology is well established for off-line analytic systems, while the application for process control strategies, like the dynamic loading of a continuous SMB chromatography, requires on- or at-line measurements.

In this study, a novel BLI-based prototype was developed, enabling continuous at-line measurements for mAb breakthrough detection in a continuous SMB chromatography process. The presented prototype was implemented in an advanced control strategy for a continuous, automated mAb manufacturing process that comprises a perfusion cultivation and a SMB capture step. The respective equipment control software infrastructure has been implemented using a visual programming approach as an enabler for a rapid and intuitive software prototyping process.

2 | MATERIALS AND METHODS

2.1 | Perfusion cultivation

A 2 L Univessel® Glass bioreactor (Sartorius) was inoculated at 0.2×10^6 cells/mL from standard batch seed cultures of a Cellca CHO DG44 cell line (Sartorius) expressing a mAb of the type IgG1. The process was controlled at 36.8°C, pH 7.1 and DO 60%. After a 3-day batch phase, pH was shifted to 6.95 and perfusion was started at 1 vessel volume per day (VVD) using a proprietary perfusion media formulation. An ATF2 device (Repligen) was connected to the bioreactor bottom drain for cell retention. During the initial cell growth phase, the perfusion rate was increased as required to maintain a cell specific perfusion rate (CSPR) of 50 pL/(cell × day). In summary, perfusion rate was controlled using a gravimetric feed flow

controller in combination with removal of cell-free permeate to maintain a constant working volume of 2 L. An automatic cell bleed was utilized to maintain the target viable cell concentration (VCC). This was achieved using a PID controller in BioPAT® MFCS (Sartorius) to control the bleed pump speed based on on-line biomass measurement (BioPAT® ViaMass; Sartorius). As control off-line samples were taken manually and the VCC as well as the viability were measured with a CedexHiRes Cell Counter (Roche).

2.2 | Continuous capture

The cell free perfusion permeate was transferred to a 2 L surge tank and from there continuously processed by a Resolute® BioSMB PD chromatography system (software version 14.06; Sartorius). Three 5 mL MabSelect SuRe™ pre-packed HiTrap columns (2.5 cm column height, 1.6 cm column diameter; Cytiva) were used for the affinity mAb capture. The initial recipe for all phases was: loading with the flow-through of the previous column in the loading cycle, loading with the perfusion permeate, 3 column volumes (CV) interconnected wash, 8.5 CV wash, 4 CV elution, 5 CV equilibration, 5 CV cleaning in place (CIP), and 5 CV equilibration. One cycle was completed when all three columns have finished these steps. All used buffer chemicals for equilibration and wash (PBS, pH 7.4), elution (50 mM $C_2H_3NaO_2$, pH 3.0) as well as CIP (0.1 M NaOH) were purchased from Carl Roth. The flow rate for all non-loading steps was kept constant at 39 cm/h (1.3 mL/min).

For the advanced control strategy, the dynamic flow control as well as dynamic loading function of the Resolute® BioSMB PD system were used with the respective OPC UA interface. For the dynamic flow control three phases with different loading flow rates (low = 63 cm/h [2.1 mL/min], medium = 72 cm/h [2.4 mL/min], and high = 81 cm/h [2.7 mL/min]) were programmed. All of them were based on the same recipe (see previous section) with equal number

and relative duration of each step to allow a change between the phases during operation. For the dynamic loading function, the loading of perfusion permeate was performed for at least 52 CV. This was the predefined minimum loading volume corresponding to the highest expected mAb titer of 1 g/L throughout the continuous USP. This minimum loading volume and the resulting minimum loading time also ensured that the column that was not currently being loaded could complete all non-loading steps before the column switch. The loading was continued until mAb was detected in the flow-through of the last column in the loading cycle while the column after completion of the non-loading steps waited for the column switch to be triggered.

2.3 | Continuous biolayer interferometry

The mAb breakthrough of the SMB capture process was specifically detected in the flow-through by a continuous BLI prototype (Figure 2) using a biosensor with immobilized protein A (Octet ProA biosensor; Sartorius) and a customized software.

One measurement cycle consisted of the following steps: sample application (step number 1; flow-through of the last sequential connected column during the loading step), regeneration (step number 2; 10 mM glycine, adjusted with HCl to pH 2.0), and neutralization (step number 3; PBS, pH 7.4). A complete cycle lasted 60 s (Figure 3, left). The different solutions required for each of the steps were provided by the use of a valve upstream and a pump downstream of the biosensor (Figure 2). For all steps the pump rate was 120 rpm resulting in a flow rate of 1.09 mL/min. The biosensor was illuminated by a lamp with white light and the interference of the reflected light from an internal coating within the biosensor and the end of the probe was detected by a spectrometer (Figure 2, right). According to the principle of BLI (Kumaraswamy & Tobias, 2015; Müller-Esparza et al., 2020) immobilization of the mAb to the protein A biosensor resulted in an increase of the thickness at the end of the probe resulting in an increased wavelength shift and binding signal (Figure 3, left). This is proportional to the mAb concentration in the solution (Figure 3, right).

To obtain a concentration dependent signal with low lag time the binding rate of step number 1, corresponding to the first derivative of the binding signal was used. In Figure 3, left an exemplary course of the binding signal and the binding rate is shown for the three steps for three consecutive cycles. From this it becomes apparent, that the maximum binding rate value is obtained more rapidly compared to the maximum binding signal, which reaches a maximum plateau only after a prolonged period of time (Figure 3, left). The maximum binding rate exhibits an explicit, nonlinear correlation with the mAb concentration (Figure 3, right).

2.4 | Control strategy architecture

The process control component used to orchestrate the instruments embedded into the continuous bioprocess was implemented using the Node-RED integrated development environment (Version 1.3.4; OpenJS Foundation). All devices were part of the same network infrastructure. The communication nodes to interact with the laboratory equipment were “node-red-contrib-opcua” for the interaction with the OPC UA interfaces of the Resolute® BioSMB PD chromatography system and the permeate surge tank balance (Cubis II; Sartorius) implemented into BioPAT® MFCS. In addition, the inbuilt TCP node for the interaction with the continuous BLI prototype software (all Sartorius) was used. Timer related events were implemented using the “node-red-contrib-looptimer-advanced” node. Furthermore, the inbuilt core nodes were used to establish the control script. The software component was deployed on a Raspberry Pi 4 B 4GB single-board computer running on Raspberry Pi OS (Debian 11; Kernel 5.15; Raspberry Pi Foundation).

2.5 | Analytics

For the determination of mAb concentration size exclusion chromatography (SEC) was used as high performance liquid chromatography (HPLC) with a Dionex UltiMate 3000 HPLC System (Thermo Fisher Scientific) and a Yarra 3 µm SEC 3000 column (Phenomenex) at a flow

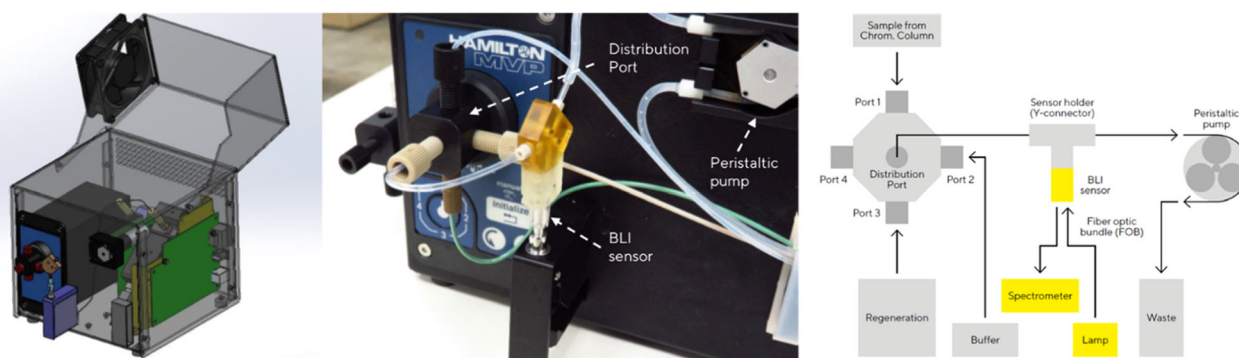


FIGURE 2 Setup of the continuous BLI prototype. A schematic representation of the entire prototype (left), a picture exhibiting the distribution port, BLI sensor, and peristaltic pump (middle) as well as a representation of the flow paths (right) are shown. BLI, biolayer interferometry.

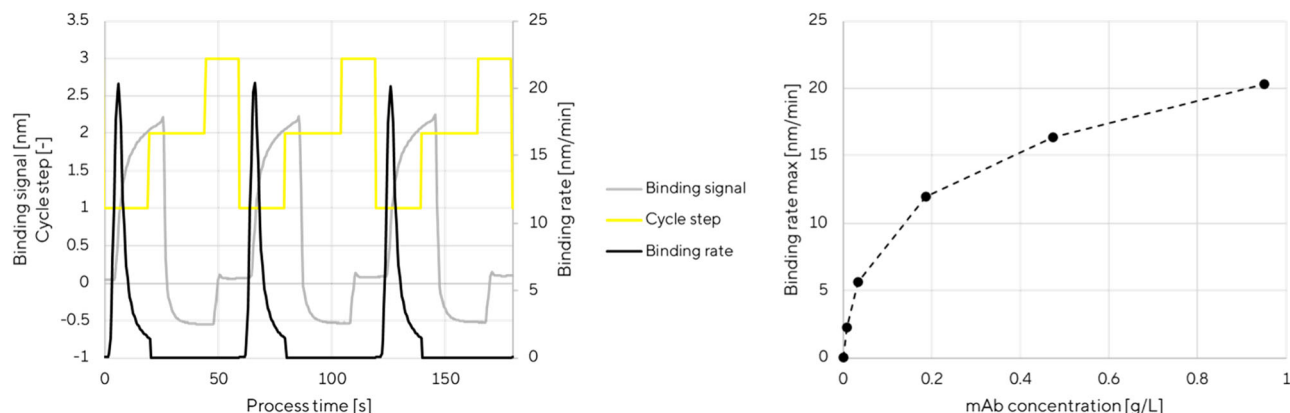


FIGURE 3 Exemplary binding signal and binding rate for three cycles of continuous BLI measurement of perfusion permeate (left). Indicated cycle steps are the sample (1), regeneration (2), and equilibration (3) step. A nonlinear correlation between the concentration of purified mAb and the maximum binding rate was obtained for purified mAb samples diluted in PBS (right). BLI, biolayer interferometry; mAb, monoclonal antibodies.

rate of 1 mL/min. A SEC buffer solution containing 0.1 M Na_2SO_4 , 0.05 M NaH_2PO_4 , and 0.05 M Na_2HPO_4 (all chemicals purchased from Carl Roth) with a final pH of 6.6 was used. The SEC method was previously verified with analytical protein A HPLC to ensure appropriate mAb quantification (data not shown). If necessary, samples were diluted with SEC buffer before analysis and filtered through a Minisart RC4 0.2 μm syringe filter (Sartorius). The mAb concentration was determined based on the peak area (at 220 nm) of a calibration curve obtained using a known reference mAb material.

The HCP concentration was determined by a CHO HCP-ELISA (Cygnus Technologies) according to the manufacturer's instructions. Samples were diluted, if necessary, in ELISA buffer (20 mM TRIS, 50 mM NaCl; all chemicals purchased from Carl Roth). Measurement was performed with an Infinite M Nano Plus plate reader (Tecan Trading AG) with a limit of detection of 1 ng/mL.

Concentration of DNA was measured by the Quant-iT™ Pico-Green™ dsDNA Assay Kit (Thermo Fisher Scientific). Samples were diluted in TE buffer (10 mM TRIS, 1 mM EDTA, 0.1% SDS; all chemicals purchased from Carl Roth) if necessary. The measurement was performed by an Infinite M Nano Plus plate reader with a limit of detection of 31.25 ng/mL.

The parts per million (ppm) value of HCP and DNA (ppm_{imp}) was calculated according to Equation 1, where c_{imp} is the concentration of the respective impurity and c_{mAb} is the concentration of the mAb.

$$\text{ppm}_{\text{imp}} = \left(\frac{c_{\text{imp}}}{c_{\text{mAb}}} \right) \times 1000000 \quad (1)$$

3 | RESULTS AND DISCUSSION

3.1 | Continuous USP perfusion cultivation

The perfusion process and the main control loops were successfully integrated to enable a fully automated and robust process even at

high cell concentrations (see schematic overview in Figure 4). In contrast to previously reported similar process setups (Müller et al., 2022), the two main control loops for perfusion operation, the perfusion feed and perfusion permeate, have been interchanged. This change allowed for improved process robustness and ensured optimal supply of nutrients. This was required to obtain high cell concentrations with high viabilities, by utilizing a gravimetric flow controller to maintain a steady addition of fresh perfusion media. Due to a direct coupling of the permeate controller to the bioreactor weight a constant working volume was maintained. As the working volume is not only affected by the addition of feed but also current bleeding rate as well as addition of corrective agents, for example, for foam or pH control, this resulted in the permeate flow rate being more unpredictable and susceptible to fluctuations. To compensate for the resulting fluctuations in permeate flow rate and to enable a robust, integrated process, a dynamic flow rate regulation was implemented as part of the advanced DSP control strategy.

The cell growth and viability of the perfusion cell cultivation is shown in Figure 5. As a result of the consistent cultivation conditions by the applied control loops, a high viability, and similar fast cell growth as previously reported for this cell line (Graf et al., 2022; Müller et al., 2022; Schulze et al., 2021) was observed during the initial growth phase. An automated cell bleed was started on Day 5 just before the target VCC of 45×10^6 cells/mL was obtained, ensuring a smooth transition to the steady-state like phase of the process (Figure 5). The target VCC could be well controlled for the remaining perfusion operation with only one noticeable deviation on Day 10. Here the actual VCC was about 30% below the target VCC (Figure 5), which resulted from changes in the VCC-capacitance-correlation and thus deviations of on-line and off-line VCC measurements. Simultaneously the mAb concentration in the permeate stream decreased due to the lower VCC (Figure 5). The process stabilized quickly after performing a recalibration, demonstrating the capability of the applied perfusion control strategy to maintain steady cultivation conditions. The continuous product capture from the

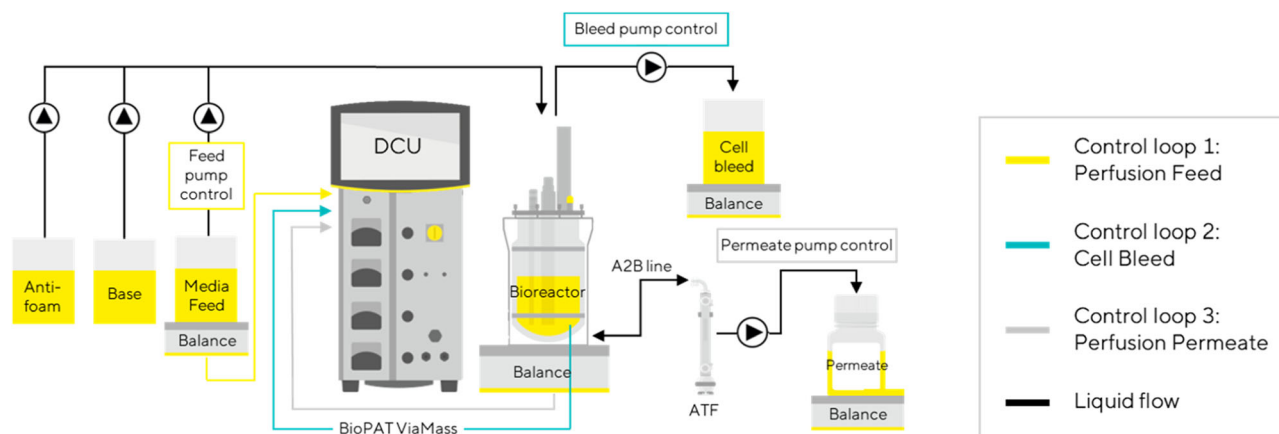


FIGURE 4 Schematic representation of the USP perfusion process with indicated control loops. USP, upstream process.

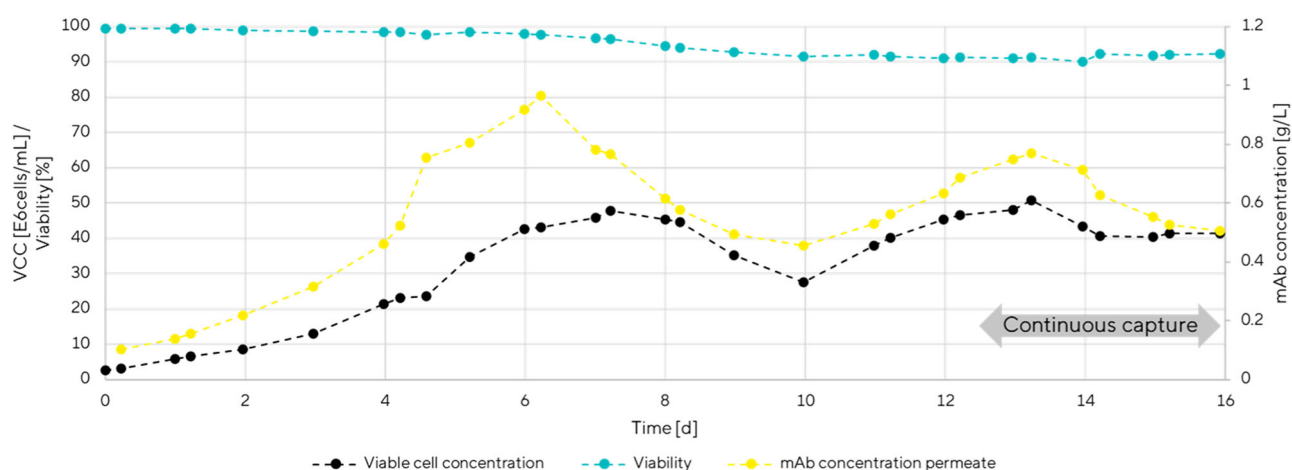


FIGURE 5 Cell growth and viability of the perfusion cultivation in the 2 L bioreactor and the mAb concentration in the permeate stream. An automated cell bleed using an on-line bio-capacitance probe was started on Day 5, just before the target VCC of 45×10^6 cells/mL was obtained. mAb, monoclonal antibodies; VCC, viable cell concentration.

perfusion permeate was implemented after the system stabilized, between cultivation Days 12 and 16. During this period the average mAb concentration in the permeate stream was approximately 0.7 g/L (Figure 5).

3.2 | Advanced control strategies for dynamic flow and dynamic loading of the continuous capture

For a direct continuous capture of the mAb from the perfusion cultivation a continuous affinity SMB chromatography was used. To establish a robust process and to be able to respond to fluctuations of the USP an overarching advanced control strategy was developed.

The successful integration of the USP perfusion and the DSP capture step is schematically shown in Figures 4 and 6. The cell free perfusion permeate was collected in a surge tank placed on a balance. The surge tank was in turn directly connected to the load inlet of the SMB chromatography system. An OPC UA interface was used to

transmit the balance signal. For the dynamic flow control, which should compensate for variations in perfusion flow rate from the USP, the loading flow rate of the SMB chromatography system was adjusted to maintain the weight of the surge tank as constant as possible (approximately 1 kg) to maintain half the volume. The advanced control strategy constantly retrieved the surge tank weight as well as the current loading flow rate. Surge tank weights less than 0.9 kg resulted in an automatic switch from the medium loading flow rate to the low flow rate, while weights above 1.1 kg triggered the high flow rate (Figures 6 and 7). This enabled the volume changes to be within a certain range, thus preventing the intermediate tank from running empty or overflowing.

To compensate for titer variations within the USP and the different breakthrough times as a result of the different loading flow rates, a dynamic loading function was implemented in the advanced control strategy as well. Based on the initial chromatography recipe the loading as well as the sequential connection to transfer the flow-through of one column to another, was continued until a mAb

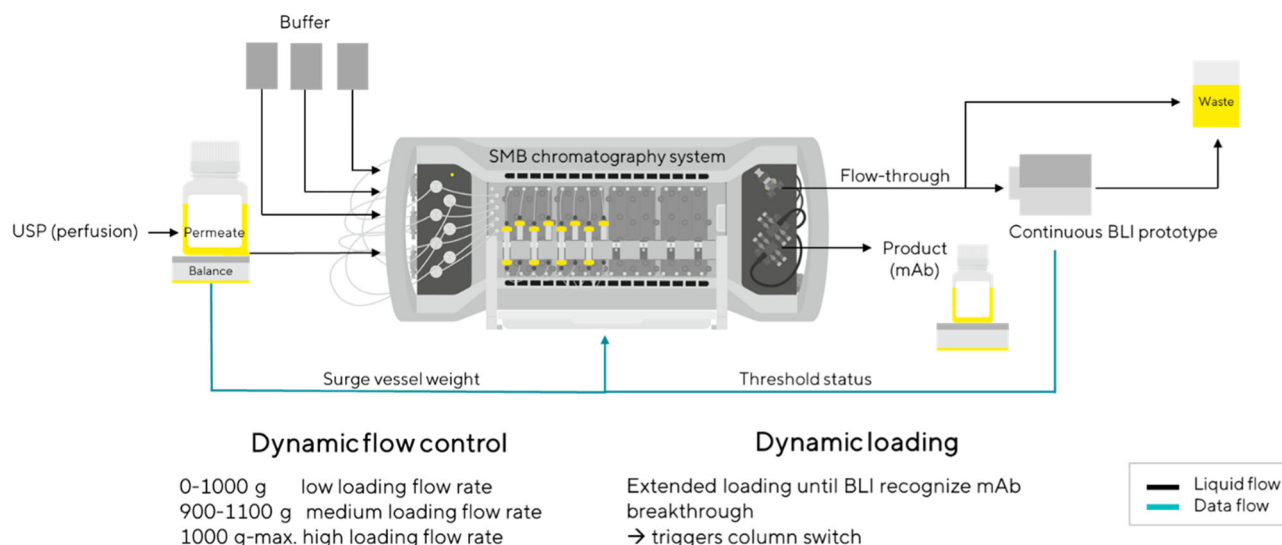


FIGURE 6 Schematic representation of the advanced control strategies for dynamic flow and loading of the continuous capture.

breakthrough was detected by the continuous BLI prototype in the flow-through of the last column in the loading cycle (Figure 6). To define the mAb breakthrough, a threshold value of the binding rate measured by the continuous BLI prototype was specified. When this value was exceeded, a signal for the column switch was sent to the SMB chromatography system via the corresponding interfaces. Higher mAb concentrations in the perfusion permeate would result in an earlier mAb breakthrough. Therefore, the threshold would be reached earlier resulting in a shorter loading time. However, the threshold value at which the column switch must be triggered remains the same since it corresponds to the same mAb concentration indicating breakthrough to the flow-through. This setup allowed that independent of the USP titer, the detection of low mAb concentrations in the flow-through (initial mAb breakthrough) could be used to perform a dynamic loading.

Very low threshold values could result in a false premature breakthrough detection of the mAb even for small fluctuations of the baseline signal, while high values would result in a certain loss of product in the flow-through. For the determination of the threshold value the flow-through at the beginning of the loading step (representing a blank value without mAb breakthrough; approximately 0.4 nm/min) and a 20-fold diluted sample of the perfusion permeate (representing a maximum mAb breakthrough of 5%; approximately 8 nm/min) were previously measured. Within this range, a low threshold value of 1 nm/min was selected as a trade-off to ensure robust mAb breakthrough detection, that is distinguishable from the baseline signal, while avoiding product loss due to column overload.

After breakthrough detection the column switch was triggered and the continuous BLI measurement was in wait mode for 108 min. This corresponds to the minimum loading time of the SMB chromatography based on the highest expected mAb titer throughout the USP to avoid unnecessary measurements. If the maximum expected mAb concentration in the perfusion permeate is exceeded,

or a loss of binding capacity occurs during the continuous capture, it may be necessary to reduce the minimum loading time and accordingly the waiting time to avoid that a very early mAb breakthrough is not detected by the continuous BLI prototype. Otherwise, this would result in a column switch being triggered immediately after the start of the measurement what could be too late and cause potential loss of product.

To obtain a synchronization between the SMB chromatography system and the continuous BLI prototype the status of the chromatography system was continuously polled by the advanced control strategy software component. A pause/resume functionality of the overall control script and the associated waiting time of the continuous BLI prototype was implemented. Thus, a stable process was ensured even in the case of unexpected events, such as potential overpressure in the chromatography system, which would interrupt the process (Figure 7).

Throughout the continuous capture in total 34 loading steps were performed by the SMB chromatography system within approximately 3 days (Figure 8). To start the process and for an initial loading of the columns, the first three loading steps were performed according to the predefined initial recipe without a dynamic control strategy. Accordingly, the loading times for loading steps 1–3 were exactly 108 min which was the loading time of the initial recipe and corresponds to the highest expected mAb titer of 1 g/L throughout the USP (Figure 8). Based on the dynamic loading strategy presented here, there could be additional benefits for the start-up and shut-down cycle to further optimize resin utilization throughout the entire process. However, this was not part of the study presented here.

Beginning with the second cycle, corresponding to the fourth loading step, the continuous BLI measurement was automatically started by the advanced control strategy after 108 min loading time. The flow-through of the last (in this setup the second) column in series during the interconnected loading step was monitored for mAb

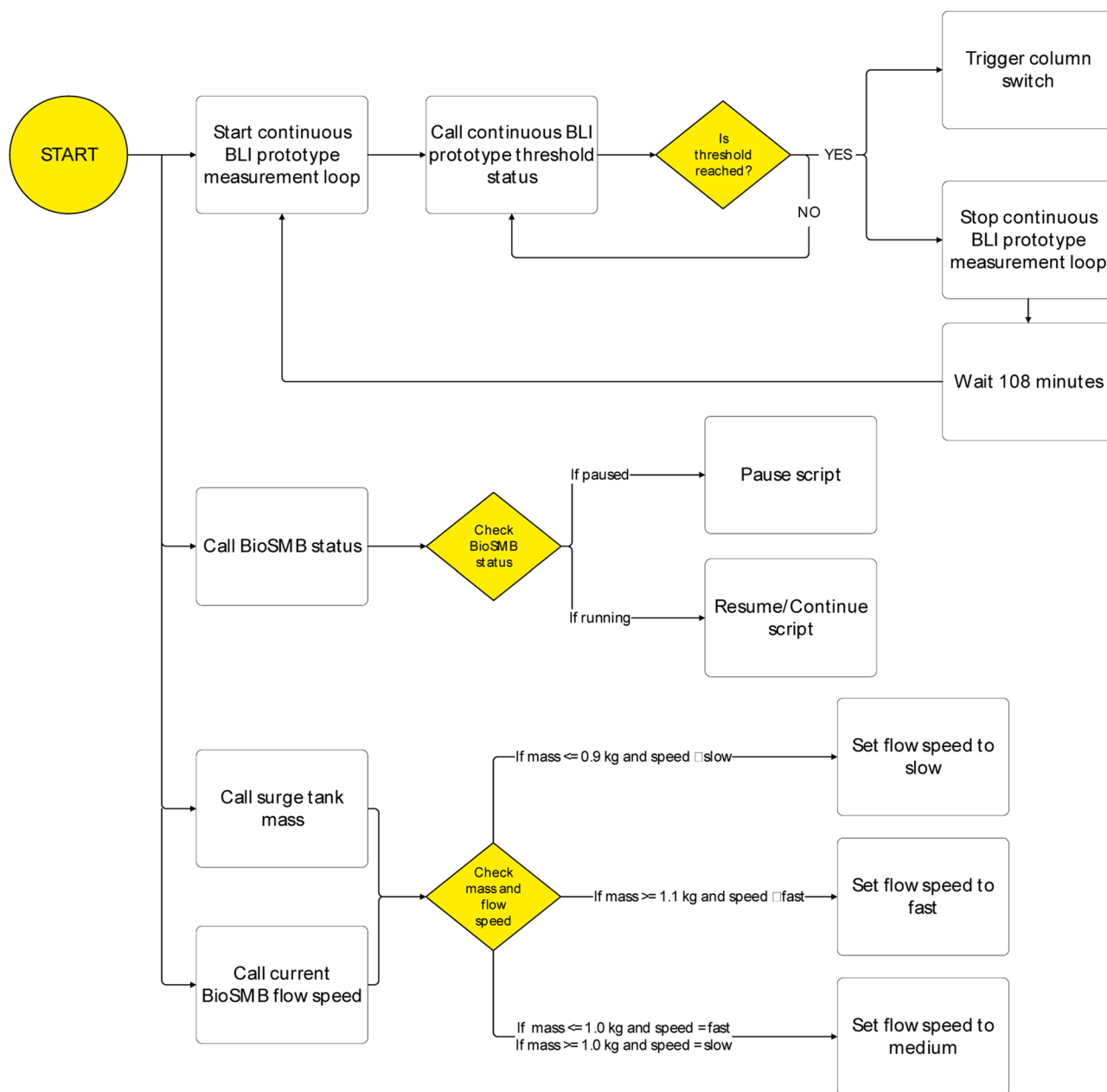


FIGURE 7 Advanced control strategy script for the DSP part of the continuous bioprocess. DSP, downstream process.

breakthrough. In Figure 8 (top) the binding rate signal from the continuous BLI prototype is shown exemplary for the seventh loading step. Due to the cyclic measurement principle a binding rate signal was obtained every minute. The binding rate remained at a low baseline level below 0.4 nm/min until approximately 121 min loading time indicating the absence of significant amounts of mAb in the flow-through (Figure 8). From that time on the value continued to increase representing a gradual breakthrough of mAb from the column. At the 27th cycle of the continuous BLI measurement (starting after 134 min loading time), the predefined threshold of the binding rate of 1 nm/min was exceeded (Figure 8), resulting in a stop of the measurement and a column switch of the SMB chromatography by the advanced control strategy (Figure 7). The mAb

concentration in the flow-through was determined exemplary at the end of the 7th and 27th loading step (just before the threshold was reached) to be below 0.02 g/L, indicating no significant product loss due to the dynamic loading. Using the continuous BLI prototype, it was thus possible to perform a robust and specific mAb breakthrough detection with low background signal.

Throughout the entire process mAb breakthrough was only observed after 108 min (Figure 8, bottom), which confirmed the calculated shortest expected loading time. The loading time for the fourth loading step was significantly longer compared to the other loading steps due to an underloading of the columns within the initial start cycle (first three loading steps) of the SMB chromatography, which was not controlled by the dynamic loading strategy. The

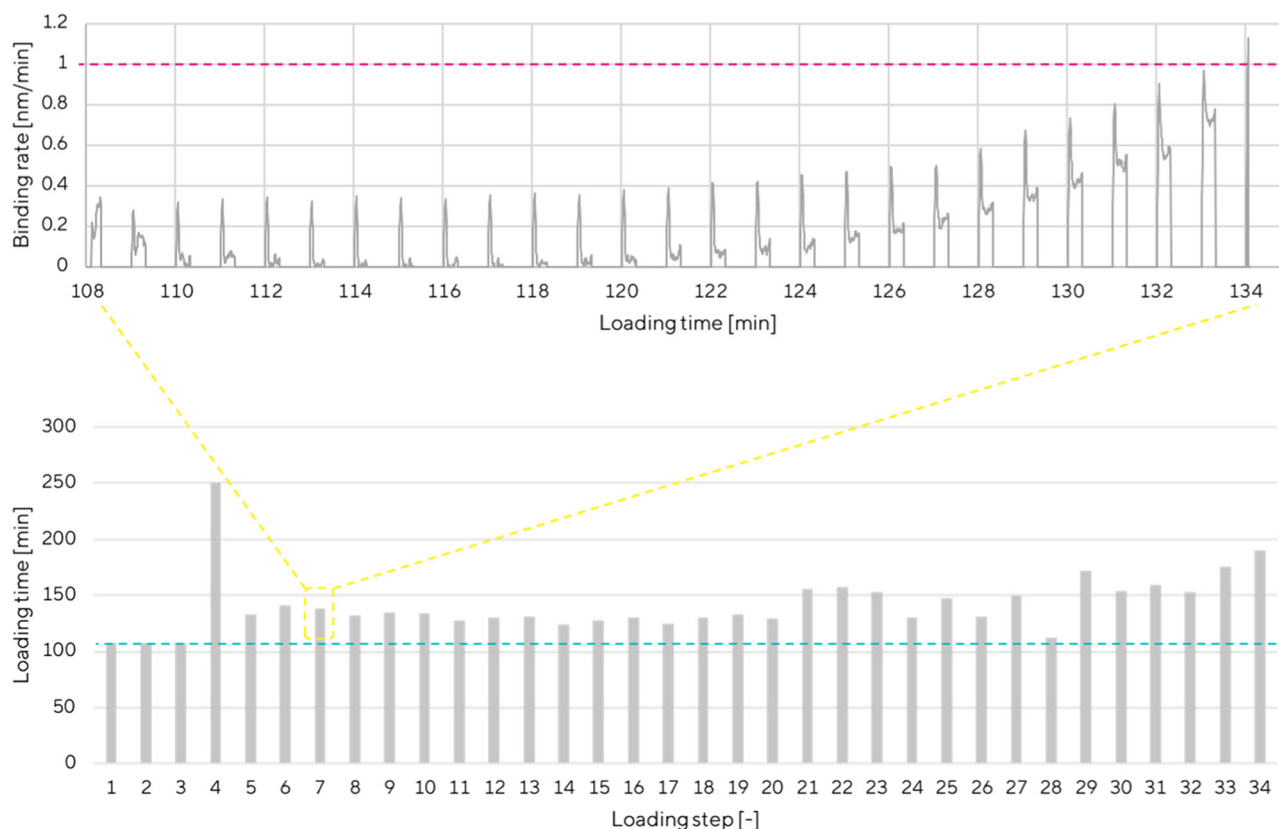


FIGURE 8 Top: Binding rate of the continuous BLI prototype throughout the measurement of the seventh loading step until the predefined threshold (1 nm/min, dashed magenta line) was reached and the column switch of the SMB chromatography system was triggered. Bottom: Loading time of the SMB chromatography system for each of the loading steps. The standby time of the continuous BLI prototype, where no measurement during the loading step was performed, is indicated (108 min, turquoise dashed line). BLI, biolayer interferometry; SMB, simulated moving bed.

TABLE 1 Concentration of the mAb throughout the three process days for the permeate as well as the different chromatography outlet pools.

Sample	mAb concentration Day 1 [g/L]	mAb concentration Day 2 [g/L]	mAb concentration Day 3 [g/L]
Permeate	0.81	0.76	0.55
Eluate	14.5	16.4	14.4
Flow-through	0.013	0.013	0.013
Wash	0.080	0.164	0.013
Waste	0.013	0.014	0.013

Abbreviation: mAb, monoclonal antibodies.

average loading time from loading step 5–20 was approximately 130 min. From the 20th loading step to the end of the process, longer loading times occurred with an average of approximately 150 min. This was due to a decreased mAb titer in the permeate (Table 1) and because the loading flow rate was partly reduced by the dynamic flow control in this time range (Figure 9), resulting in a later mAb breakthrough (Figure 8, bottom). The additional loading time gained by the dynamic loading approach in comparison to a conservative

predefined recipe (Figure 8, loading times above the turquoise dashed line) directly correlates to an additional mass of mAb which can be processed per loading step. This allows to process an increased amount of mAb for a given resin lifecycle and demonstrates a maximal resin utilization.

After the initial filling of the surge tank at the beginning of the process the first cycle, corresponding to the first three loading steps, were performed based on the predefined recipe with the medium loading flow rate (Figure 9). For the next cycle, starting from 5.4 h process time, the dynamic flow control strategy has taken effect, resulting in a switch to the fast flow rate after the threshold of 1.1 kg was exceeded after approximately 9 h (Figure 9). However, due to the slightly increased VCC at the beginning of the continuous capture integration (Figure 5) and the consequently higher permeate flow rate, the level of the surge tank further increased up to 1.2 kg. However, after the VCC had readjusted back to the target value of approximately 45×10^6 cells/mL, and maintaining the CSPR, the fast loading flow rate was sufficient to reduce the surge tank weight back to the desired range. After approximately 45 h process time the weight dropped below 1 kg resulting in a switch to the medium flow rate and a subsequent increase of the surge tank level (Figure 9). Throughout the following process time the surge tank weight varied

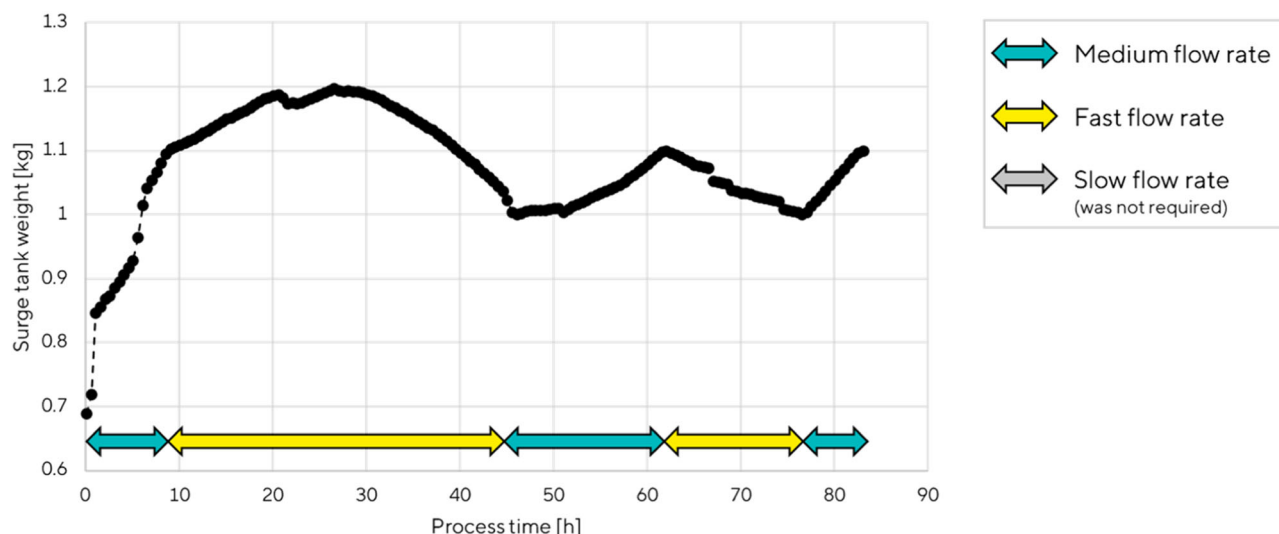


FIGURE 9 Surge tank weight throughout the continuous capture. For the given process only the medium and fast flow rate were required to maintain the desired surge tank weight.

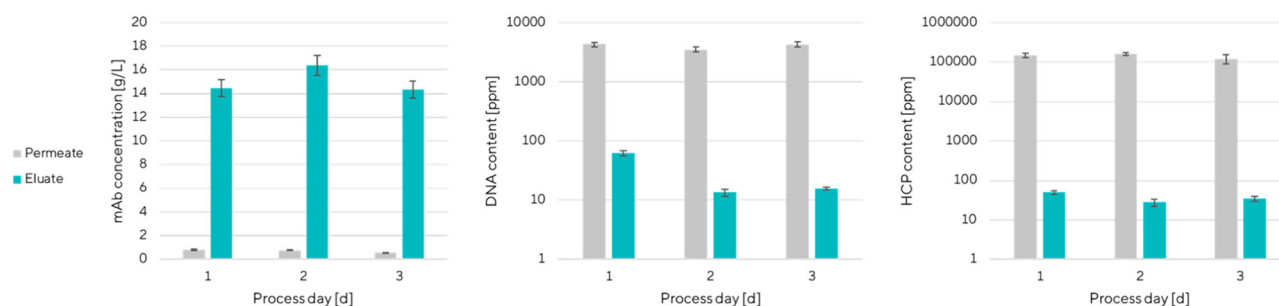


FIGURE 10 Analytical results for the permeate and eluate. The mAb concentration (left) as well as the DNA (middle) and HCP (right) content are shown. mAb, monoclonal antibodies.

between 1 and 1.1 kg which was within the desired range according to the control strategy (Figure 9).

The different outlets of the chromatography system were pooled each for every process day and analyzed together with a permeate sample from the surge tank (Table 1 and Figure 10). Throughout the 3 process days the mAb concentration in the permeate, which was the feed for the continuous SMB chromatography, decreased from 0.81 to 0.55 g/L. This was probably due to a slight decrease of the VCC throughout this time (Figure 5) and product sieving effects during the USP perfusion process (Clincke et al., 2013; Gagnon et al., 2018). However, in the eluate pool a high mAb concentration between 14.4 and 16.4 g/L with a mean value of 15.1 g/L was obtained. Potential reasons for the differences could be the initial start-up cycle on Day 1 (was performed without the dynamic loading function), slight changes in the binding capacities of the columns throughout the process (elution was performed with a fixed volume, which results in a change in the concentration of the eluate when the capacity changes) or differences between the individual columns (sampling was performed once a day and not after passing a certain number of cycles). However, the mean mAb concentration in the eluate was significantly higher compared to a

process based on a similar setup but with a conservative predefined SMB chromatography recipe, where a mAb concentration of approximately 7 g/L was obtained (Müller et al., 2022). This again demonstrates a maximal resin utilization by the dynamic loading control strategy, despite variations of the product concentration within the USP.

Throughout the entire process only traces of mAb (0.013 g/L, representing the limit of detection of the SEC used for quantification) were detected in the flow-through, again confirming no significant product loss by the control strategy. While also the pooled waste fractions exhibit no significant mAb concentration, small amounts of the product were detected in the wash fractions, probably due to a too short interconnected column wash step.

Because of the constantly changing mAb concentration in the perfusion permeate (shown exemplary for each individual process day in Table 1) and the variable loading flow rate (Figure 9), the mass input of mAb for the calculation of the yield could not be determined exactly. For this reason, the yield was calculated as proportion of the mAb mass in the eluate fraction related to the total mass of mAb recovered in all outlets of the chromatography system. Thus, for the complete process a high mAb yield of 96.9% was obtained.

Simultaneously, process related impurities were significantly removed throughout the process. DNA was removed to an average amount of 30.3 ppm, representing a log reduction of 2.1 (Figure 10). Moreover, a log reduction of 3.6 was obtained for HCP resulting in an average amount of 37.5 ppm in the eluate (Figure 10). The removal of process related impurities obtained in this study was even a little bit higher compared to a very similar process which was based on a predefined SMB chromatography recipe (3.2 log reduction for HCP and 1.8 log reduction for DNA related to the ppm content [Müller et al., 2022]).

4 | CONCLUSION

For the perfusion process, the dynamic control strategy of the perfusion feed, cell bleed and perfusion permeate resulted in an automated maintenance of high cell densities along with high viabilities. These dynamic adaptations and the cultivation itself, as a biological system, resulted in varying process conditions feeding into the downstream processing. These variations, such as the mAb concentration and permeate flow rate as well as fouling effects throughout the resin lifecycle or differences of the column binding capacity were addressed by the advanced control strategy.

Due to the dynamic loading control, column loading was automatically optimized to obtain maximum resin utilization without significant product loss. In addition, a robust continuous process was ensured by the dynamic flow control which maintained the filling volume of the permeate surge tank within the desired range throughout the complete process. Resulting changes of the loading flow rate, which in turn changed the breakthrough time point during the loading step, were also addressed by the dynamic loading. This further highlights the interaction and mutual benefit of the applied DSP control strategies.

The advantages become even more obvious when comparing the results of this study with a similar process without adaptive control (Müller et al., 2022). A twofold increase of the mAb concentration in the eluate was obtained by maximum resin utilization due to the dynamic loading compared to the conservative predefined recipe. This probably allows further improvements of the productivity for the subsequent DSP unit operations. At the same time a high removal of process related impurities was obtained.

An enabler of the work was the implementation of the control strategy using a visual programming environment that enabled a rapid and intuitive software development process. This highlights the suitability of this tool rapid execution of novel bioprocess automation tasks in a test bed environment.

The novel continuous BLI prototype used in this study was able to reliably detect the mAb breakthrough for every loading step. Due to the specific measurement principle only very low background signals were obtained, which might represent small amounts of leached mAb during the loading. Throughout the entire process no adjustment of the threshold or exchange of the biosensor was needed although over 1000 measurement cycles were performed.

In summary, the novel prototype for continuous BLI measurements overcomes several limitations of commonly used spectroscopy methods for the advanced control of bioprocessing unit operations such as the capture step shown in this study. In addition, this novel sensor format enables the potential expansion of the technology to other modalities (such as viruses, exosomes) and impurities such as host cell protein and glycosylation for improved yield and efficiency.

AUTHOR CONTRIBUTIONS

Thomas Kruse: Had made substantial contributions to conception and design, acquisition of data, analysis and interpretation of data; had been involved in drafting the manuscript. **Jonas Austerjost:** Had made substantial contributions to conception and design, analysis and interpretation of data; had been involved in drafting the manuscript. **Johannes Lemke:** Had made substantial contributions to conception and design, analysis and interpretation of data; had been involved in drafting the manuscript. **Yuri Krasov:** Had made substantial contributions to conception and design; had been involved in revising the manuscript. **Vasily Popov:** Had made substantial contributions to conception and design; had been involved in revising the manuscript. **David Pollard:** Had made substantial contributions to conception and design; had been involved in revising the manuscript. **Markus Kampmann:** Had made substantial contributions to conception and design, analysis and interpretation of data; had been involved in revising the manuscript.

ACKNOWLEDGMENTS

The authors would like to thank the entire Advanced BioProcessing group of Sartorius, Corporate Research, especially Janina Kilian for the excellent laboratory work. Furthermore, we would like to thank the Engineering North America group in the Product Development organization of the Lab Products & Services Division for the prototyping of the continuous BLI prototype.

CONFLICT OF INTEREST STATEMENT

The authors are employees of Sartorius and used some, but not exclusively, products of this company for this study. Patent applications are currently pending based on this work.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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How to cite this article: Kruse, T., Austerjost, J., Lemke, J., Krasov, Y., Popov, V., Pollard, D., & Kampmann, M. (2023). Advanced control strategies for continuous capture of monoclonal antibodies based upon biolayer interferometry. *Biotechnology and Bioengineering*, 1–13. <https://doi.org/10.1002/bit.28586>